

Riffat

Backend Development & AI Engineering Portfolio

[LinkedIn Profile](#)

Portfolio Framed Cases

Voice Card

Clear, honest, practical, always learning, no buzzwords.

About Me

Hi, I'm Riffat, a BSCS student at the University of Engineering and Technology (UET), Taxila.

I entered Computer Science after a pre-medical background, so learning programming meant starting from the basics. Instead of trying to learn everything at once, I focus on building small projects that teach me one concept at a time. My current interests are Backend Development, Python, SQL, Docker, and AI Engineering.

I use AI as a learning assistant to understand concepts, improve documentation, and organize my work. I still test my code, verify the results, and make the final implementation decisions myself.

Case Study 1 – BE-01: Smallest Possible Backend

Problem

My first backend assignment was to build the smallest possible backend application using FastAPI. The goal was not to create a complex application but to understand how a backend server works, how API endpoints respond with JSON, and how to run and test the application locally.

What I Did & What I Decided

I built a FastAPI application with two JSON endpoints and configured it to run locally using Uvicorn. After completing the implementation, I tested each endpoint in the browser and with curl to make sure the responses were correct.

One decision I made was to keep the project simple instead of adding unnecessary features. I wanted to fully understand the fundamentals before moving to more advanced topics. I also wrote a clear README so that anyone visiting the repository could understand the project and run it without confusion.

AI helped me clarify FastAPI concepts and improve the project documentation, but I personally verified the application and made the final implementation decisions.

Outcome

The application successfully met all assignment requirements. Through this project, I learned:

- FastAPI project structure
- Creating API endpoints
- Returning JSON responses
- Running applications with Uvicorn
- Testing APIs using browsers and curl
- Writing professional project documentation

Project Repository

[BE-01 – Smallest Possible Backend](#)

Case Study 2 – BE-04: Containerize Your Stack

Problem

After building a backend application, the next challenge was making sure it could run consistently on different systems. The objective was to package the application inside a Docker container so that the environment remained the same regardless of where it was executed.

What I Did & What I Decided

I created a Dockerfile for the FastAPI application, built a Docker image, and launched the project inside a container. After containerizing the application, I verified that the API still worked correctly.

Instead of simply copying Docker commands, I spent time understanding the purpose of each step. I wanted to know how images, containers, and environment consistency make deployment easier for developers.

AI helped explain Docker concepts, but I practiced running the commands myself and verified that the application behaved correctly inside the container.

Outcome

This project introduced me to Docker and containerization. By the end of the assignment I understood:

- Docker images
- Docker containers
- Building images
- Running containerized applications
- Why containerization improves deployment consistency

Project Repository

[BE-04 – Containerize Your Stack](#)

Case Study 3 – BE-06: Your First Background Job

Problem

The objective of this assignment was to understand background jobs. Instead of making users wait while a long-running task completed, the backend should return a response immediately and continue processing the task in the background.

What I Did & What I Decided

I implemented a background task in FastAPI and observed how the application could continue working after responding to the client.

Rather than treating background processing as something that "just works," I focused on understanding where it should be used and how it improves the user experience by preventing unnecessary waiting.

AI helped explain the workflow, but I tested the application myself and verified that the background process behaved as expected.

Outcome

This assignment helped me understand:

- Background tasks
- Asynchronous thinking
- Faster user experience
- Separating user requests from long-running work
- A common backend engineering pattern used in real applications

Project Repository

[BE-06 – Your First Background Job](#)

Professional Bio

I'm a BSCS student at UET Taxila who enjoys learning by building projects. My current focus is backend development and AI engineering. Every project in this portfolio represents a concept I learned through practice, testing, and documentation rather than simply following tutorials.

Contact / Call To Action

Thank you for taking the time to review my portfolio.

GitHub Profile

<https://github.com/Riffatktk>

Backend Portfolio

[BE-01 – Smallest Possible Backend](#)

[BE-04 – Containerize Your Stack](#)

[BE-06 – Your First Background Job](#)

LinkedIn

<https://www.linkedin.com/in/riffat-khattak-9b85193ba>

I'm always open to learning, feedback, and connecting with students and professionals who are interested in backend development and AI engineering.

Before vs After

Before (Generic AI)

I am a results-driven backend developer passionate about delivering innovative and scalable software solutions.

After (Edited in My Voice)

I'm still learning backend development, and every project teaches me something new. I use AI to help me learn and organize my work, but I test my code, verify the results, and make the final implementation decisions myself.

Reflection

These projects represent the beginning of my backend engineering journey. My goal isn't to present myself as an expert but to honestly show what I've learned, the decisions I made, and how each project helped me build a stronger understanding of backend development.